Image Deblurring

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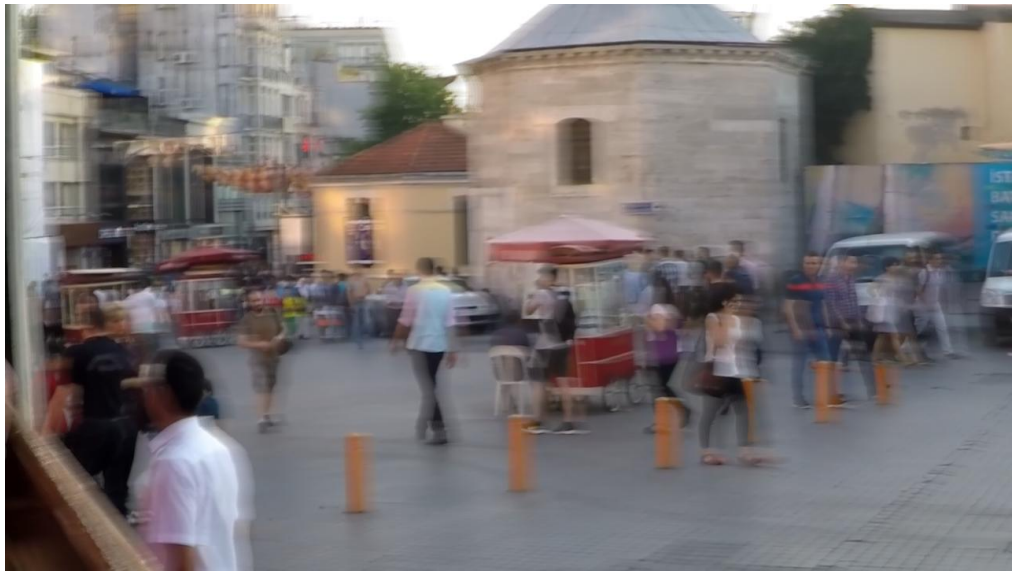
Motivation

- Defocus blur from optical diffraction
- Motion blur from camera shake

Single image deep defocus estimation and its applications - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Defocus-blur-map-refinement-a-Input-image-b-Blur-map-c-Guidance-image-d-Refined_fig4_353634777 [accessed 23 Feb 2026]

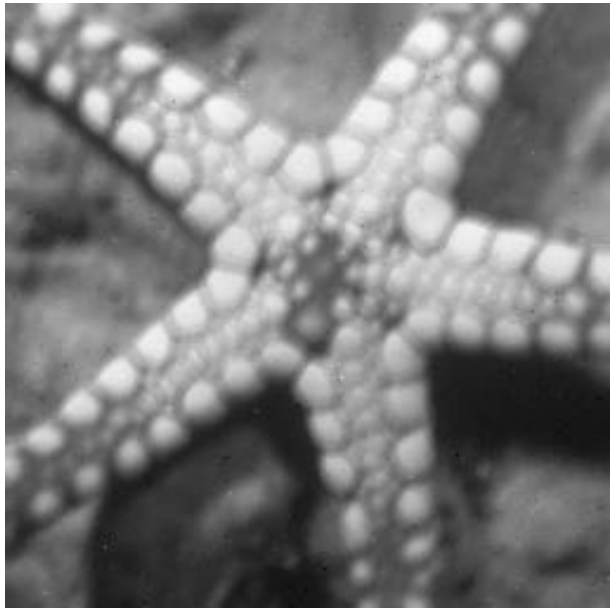


From GoPro dataset

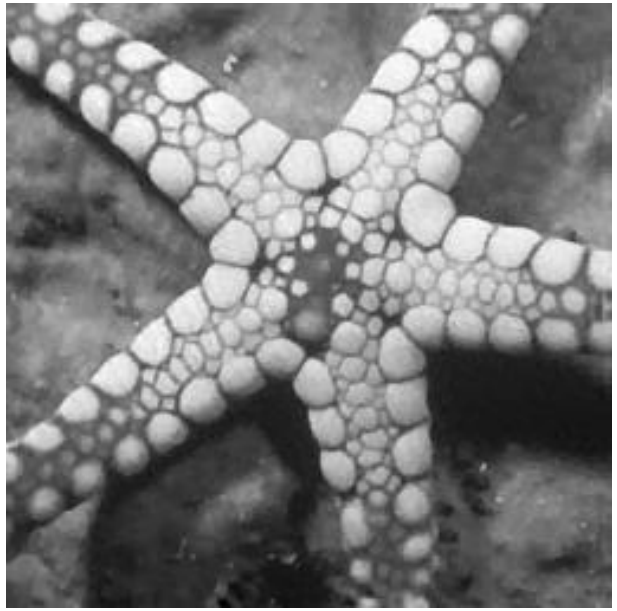
Problem Definition

- What is image deblurring?

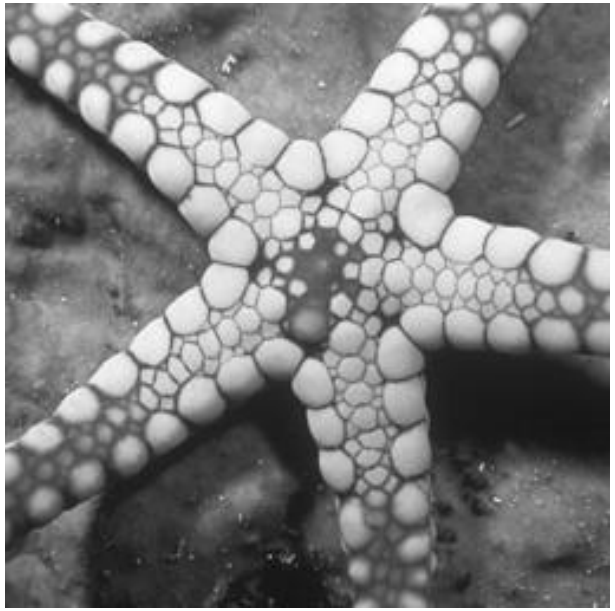
To reconstruct the sharp image from the blurred one.



Blurred Image



Restored Image



Sharp Image

Traditional method

- Blur model:

$$y = k * x + n$$

- y: Blurred image
- k: Blur kernel
- x: Sharp image
- n: Noise

- k can be seen as a coefficient map of a set of spatially shifted version of the sharp images x, and y is the linear combination of $x_{(s)}$.

Non-blind vs blind deblurring

$$y = k * x + n$$

- Non-blind deblurring

Given y and k, estimate x.

- Blind deblurring

Given y, estimate both k and x.

Non-blind deblurring

- Bayes' Theorem

$$P(x|y,k) \propto P(n) \cdot P\left(x_{g,x}\right) \cdot P(x_{g,y})$$

- The goal is to maximize $\mathrm{P}(x|y,k)$

$$\max P(x|y,k) = \min -\log P(x|y,k)$$
$$= \min ||n||^2 + \lambda\left(||x_{g,x}||^{0.8} + ||x_{g,y}||^{0.8}\right)$$
$$= \min ||y - k * x||^2 + \lambda\left(||x_{g,x}||^{0.8} + ||x_{g,y}||^{0.8}\right)$$

Solution: Fast Image Deconvolution using Hyper-Laplacian Priors

(Assumption: Noise follows Gaussian distribution and sharp image gradients follow hyper Laplacian distribution)

Limitations of traditional method

- Blind case

1. ill-posed nature lead to several possible solution pairs.
2. Definition of image priors are not robust enough to different blur.

- Spatially-variant blur

1. The blur model $y = k * x + n$ is based on uniform blur assumption, while real world blur is often Spatially-variant.

Learning-based method

- How Deep Learning overcome the limitations of traditional methods?
- Handling the “Blind case”

1. Implicit kernel learning
2. End-to-End Architecture

- Solving “Spatially-variant blur”

1. Local feature capturing

Examples

- NAFNet (2022, ECCV)

Simple backbone, non-activation function.

- Restormer (2022, CVPR)

Less computational complexity by replacing spatial attention with channel attention.

- FFTformer (2023, CVPR)

Less computational complexity by reformulating spatial attention with correlation, and thus convolution in frequency domain.

- HiDiff (2023, NIPS)

Combining Restormer and Diffusion branch.

Limitations of Learning-based method

- Quality

1. Over-smoothing: Loss of High-Frequency details induced by PSNR
2. Artifacts & Hallucinations: Fake details from generative models

- Cost

1. High Latency (Slow): Diffusion models
2. Resource Intensive (Heavy): Transformer-based models

Loss of High-Frequency induced by PSNR

- Why PSNR (L2-loss) leads to loss of High-Frequency ?

Consider:

$$x = [1, 9, 1]$$
$$y_1 = [9, 1, 1], y_2 = [1, 9, 1], y_3 = [5, 5, 1]$$

L2-loss:

$$l_1 = \sqrt{8^2 + 8^2 + 0^2} = 8\sqrt{2}$$
$$l_2 = 0$$
$$l_3 = \sqrt{4^2 + 4^2 + 0^2} = 4\sqrt{2}$$

Thus,

$$l_2 < l_3 < l_1$$

Metrics

- PSNR

$$PSNR(x, y) = 10 \cdot \log_{10}\left(\frac{MAX_y^2}{MSE(x, y)/c}\right)$$

- SSIM

$$SSIM = \frac{\left(2\mu_x\mu_y + C_1\right)(2\sigma_{xy} + C_2)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)}$$

SSIM

- $SSIM = \frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$
- For C_1 part
- Let $\mu_x = a, \mu_y = a + d$
- $\frac{(2\mu_x\mu_y+C_1)}{(\mu_x^2+\mu_y^2+C_1)} = \frac{2a^2+2ad+C_1}{2a^2+2ad+d^2+C_1} = \frac{A}{A+d^2} \leq 1$

- For C_2 part
- $\frac{(2\sigma_{xy}+C_2)}{(\sigma_x^2+\sigma_y^2+C_2)} = \frac{(\sigma_{xy}+\sigma_{xy}+C_2)}{(\sigma_x^2+\sigma_y^2+C_2)}$; $\sigma_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$

LPIPS (learned perceptual image patch similarity)

- Arxiv: The Unreasonable Effectiveness of Deep Features as a Perceptual Metric

Thank you